



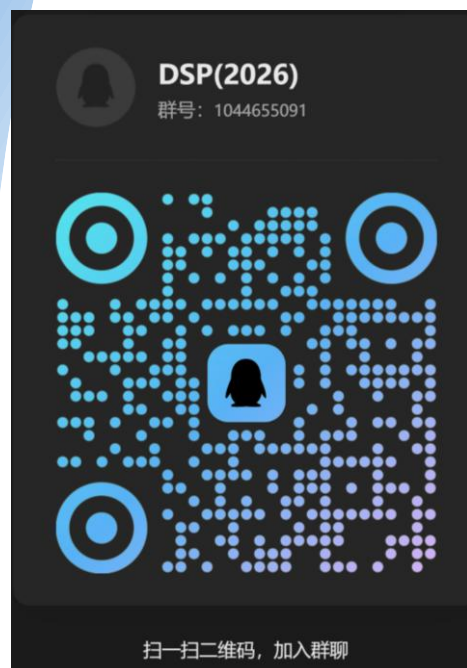
Digital Signal Processing

Lecturer : 利强

Office: 科研楼C211

Email: LQ@uestc.edu.cn

TA: 杨柳





MOOC Website:

<https://www.icourse163.org/course/UESTC-236010>

<https://www.icourse163.org/course/UESTC1-1458252163>

中国大学MOOC

课程▼

学校

数字教材

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Digital Signal Processing

播放

Galasgow college&infor. & Comm. Eng. college, UESTC

Digital Signal Processing

第7次开课 ▼

开课时间: 2025年02月25日 ~ 2025年06月30日

学时安排: 3-4小时每周

距离开课还有 7 天

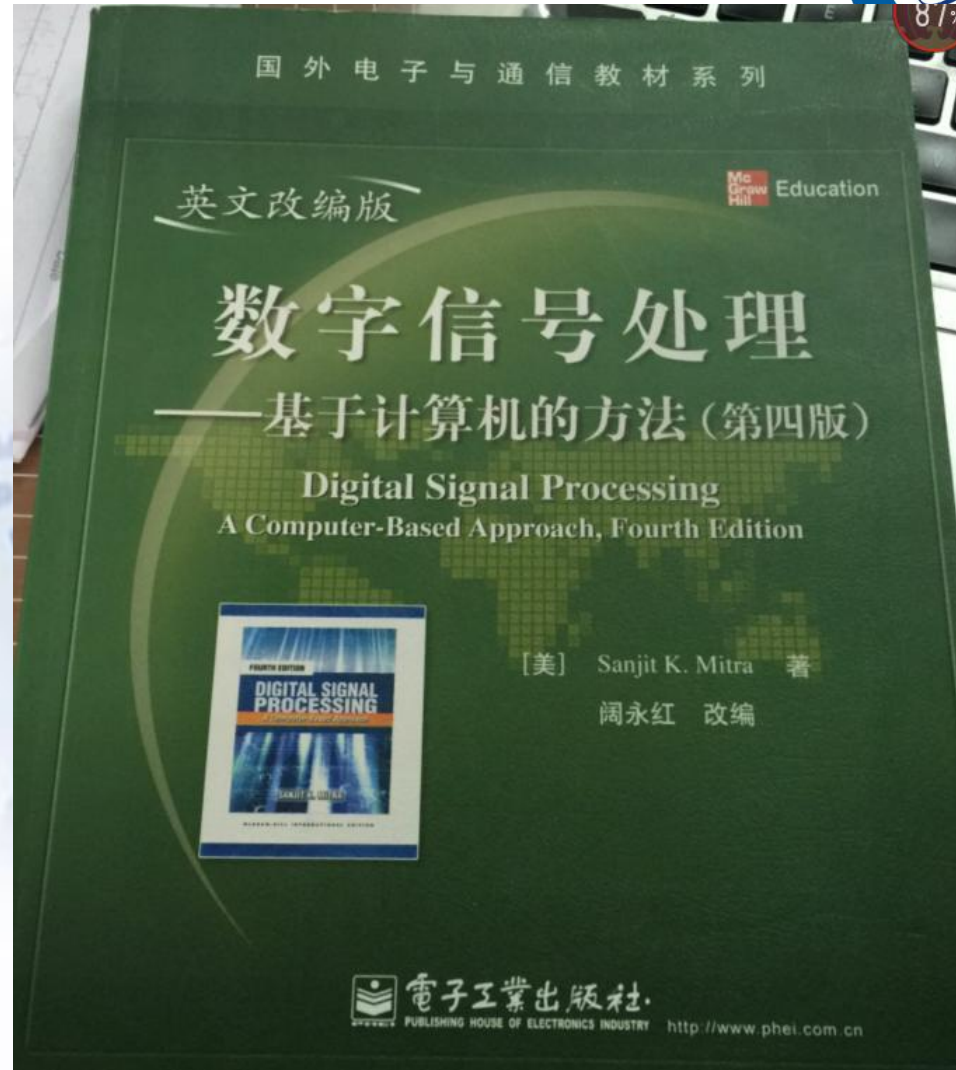
已参加, 等待开课

The Textbook



**Digital Signal
Processing---A
Computer- Based
Approach, Sanjit
K.Mitra,4th
Edition,McGraw-
Hill,2014**

电子工业出版社



Reference books



- 1. 奥本海姆（李玉柏等译），离散时间信号处理（原书第3版.精编版），机械工业出版社
- 2. 彭启琮等，数字信号处理, 高等教育出版社
- 3. 彭启琮等,DSP与实时数字信号处理,电子科大出版社
- 4. 程佩青,数字信号处理教程,清华大学出版社
- 5.胡广书，数字信号处理——理论、算法与实现，清华大学出版社

Arrangement



1. **Lectures**, 56 chs. (including course contents 48 chs. and presentations of course projects 8 chs.)
2. **MOOC**, about 32 chs.
3. **Lab time**, about 8 chs.

Grading



1. Course projects:25%

(1) Complete the design as required (Include MATLAB codes, graphs, result analysis etc.)

(2) Presentation

(3) Project report

2. Open Book Final Examination: 75%

Only one exam

Total 6 questions in 3 hours

Refer to the questions in MOOC and Homework

Contents of Mooc



Mooc videos	Book chapters
Week1: DSP-introduction and MATLAB 2videos	Ch1
Week1: Digital Processing of Continuous Signals-2	Ch2
Digital Processing of Continuous Signals-1&3	Ch3.8
Digital Processing of Continuous Signals-4	Ch12
Week2-3: DTFT 5videos	Ch3.1-3.6
Week4: Discrete Time System 4videos	Ch4
Week5-6: DFT 4videos&FFT 2video	Ch5&Ch11
Week7: ZT and IZT 5videos	Ch6
Week8-9:LTI Discrete-Time System in the Transform Domain 5videos	Ch7
Week10: Digital Filter Structures 2videos	Ch8
Week11-12: FDA tools&filter design 7videos	Ch9-10
Final Examination begins from Week 13.	

Foreword

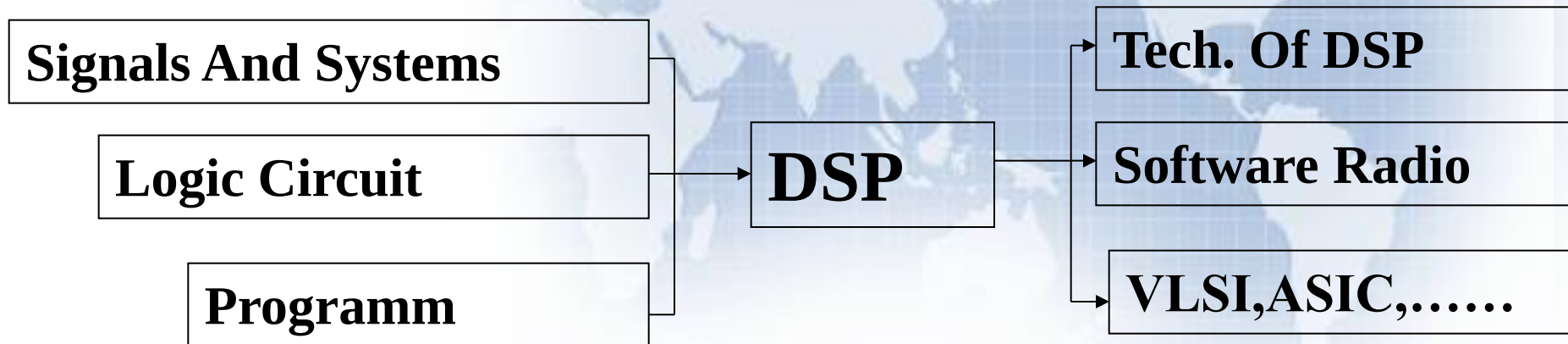


Characterization Of The Course

1.The Tech. Fields Used DSP:

Communication, Radio, Radar, Navigation, IT, etc

2. The Correlated Course:



Required Contents



- § 1
- § 3* 3.1~3.9
- § 5* 5.1~ 5.10
- § 7 7.1~7.4
- § 9 * 9.1~9.6
- § 11 11.1~11.5
- § 2 2.1~2.5
- § 4 4.1~ 4.10
- § 6 6.1~6.7
- § 8 8.1~8.6
- § 10 * 10.1,10.2,10.5
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Projects:



- **Production and Detection of Dual-Tone Multi-Frequency (DTMF) Signals**

In week 8

- **Digital Frequency Spectrum Analysis Based on DFT**

In week 10